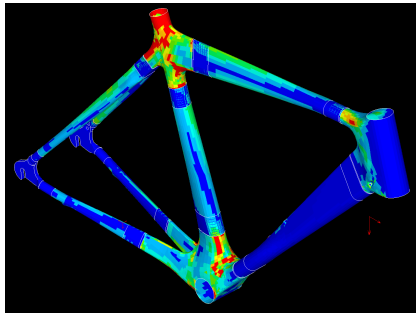
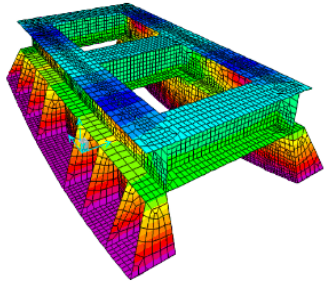


(Credit: Form Lab)



(Credit: Adesol)

Introduction

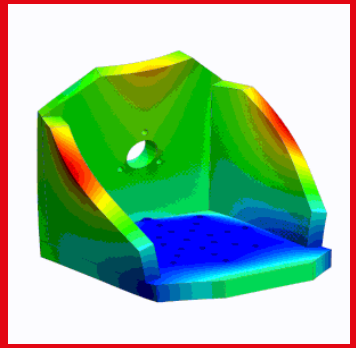
General information

ME-473

Dynamic finite element analysis of structures

Stefano Burzio

Welcome!



(Source: Sentek)

- Who are the instructors?
- How is the learning process structured?
- What reading materials are recommended?
- What does the curriculum include?
- How will my learning be assessed?
- Why is dynamic FEA important?

ME-473 Teaching staff

Lecturer



Stefano Burzio
Scientific collaborator

Teaching assistants

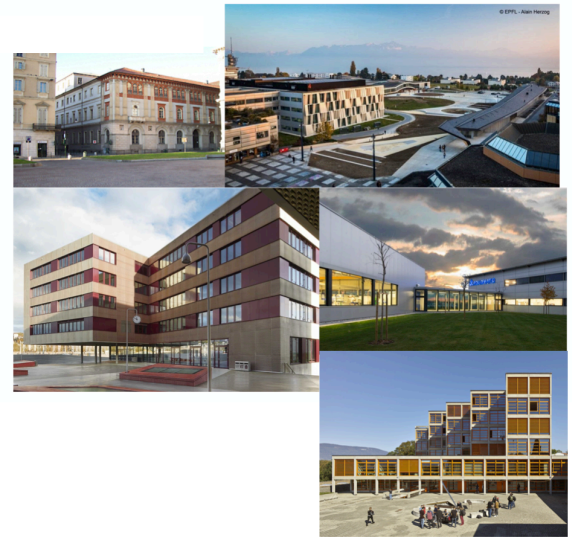


Timothée Salamon
PhD student at LFMI



Gabriel Mollet
Master's student in Mechanical Engineering

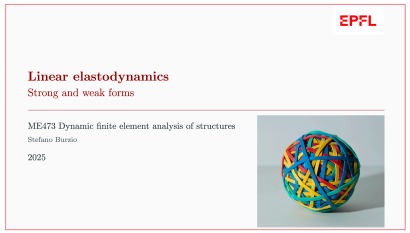
Who am I?



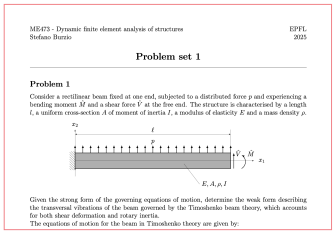
Lecture	Monday 14h - 16h	BC 01
Exercise	Monday 16h - 17h	BC 01
Office hours	Monday 11h - 12h	ME A2 390

Moodle page: (Link)

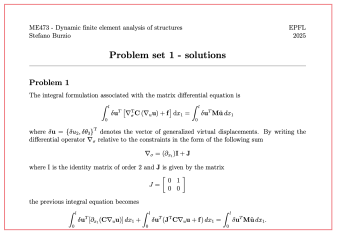
An official platform for announcements and sharing materials, including lecture notes, exercises, practice problems, and more.



Slides



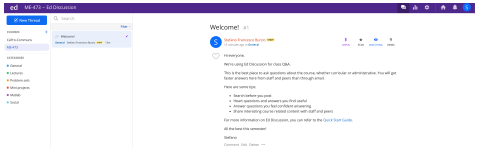
Problem sets



Problem solutions

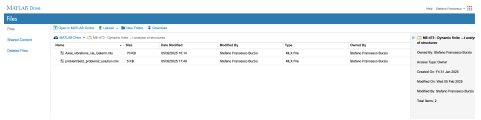
Ed discussion forum: (Link)

A general student forum, where both lecturers and fellow students can answer your questions.

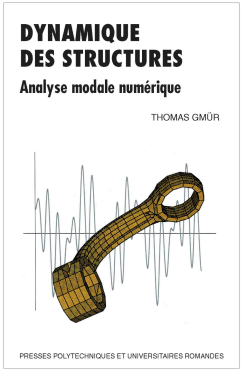


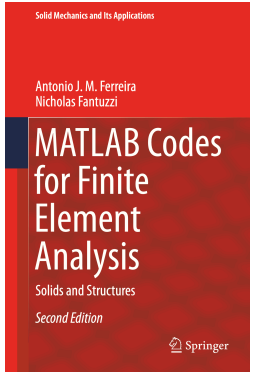
Matlab drive (Link)

Read-only online MATLAB code repository.
Log in with your Switch-edu account



Recommended books

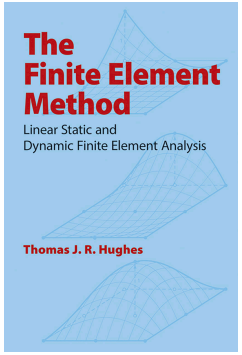


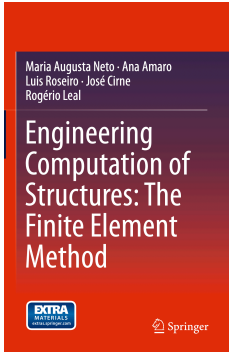


Main references:

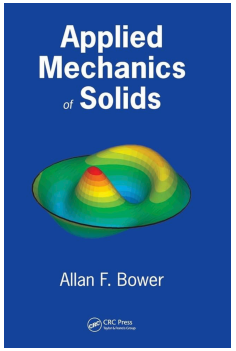
- Thomas Gmür
  Dynamique des structures (PPUR)
  EPFL library: (07 534 GMU 2012)
- Antonio Ferreira and Nicholas Fantuzzi
  MATLAB Codes for Finite Element Analysis
  Available online (link)

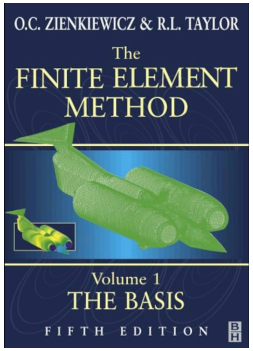
English

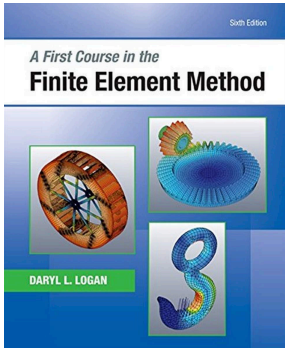




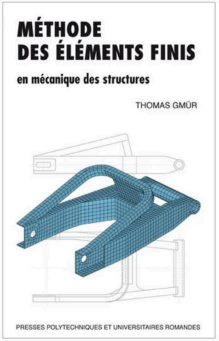
All-time classics







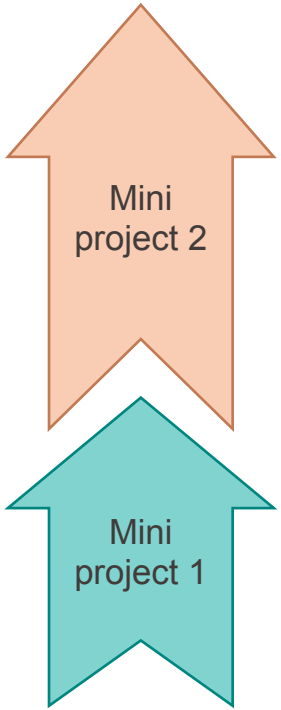
ME-372



Syllabus

Week	Module	Lecture topic	Problem set	Mini projects
1	Linear elastodynamics	Strong and weak forms	1	Groups formation
2		Galerkin and finite element methods	2	
		Jeûne fédéral		
3		Systematisation of the procedure	3	
4		Rotational or dissipative structures	4	
5	Special structural elements	Trusses	5	Project 1 statement
6		Beams, frames and grids	6	
7		Kirchoff plates	7	
8		Riessen-Mindlin plates	8	**Project 1 submission**
9	Analysis of free and forced vibrations	Lancoz, subspace algorithms	9	Project 2 statement
10		Gyroscopic and dissipative systems		
11		**CADFEM presentation : Thierry Bleuler**		
12		Newmark schemes	10	
13		Wilson schemes		**Project 2 submission**

Mini projects submission guidelines



- **Objective**: put course content into practice.
- **Groups**: 3 to 4 students (register groups on Moodle)
- **Group assessment**:
  - work divided equally among the group.
  - evaluation criteria on Moodle.
- **Deliverables**:
  - PDF report (explaining your analysis and detailing the code)
  - Simulation file (MATLAB, ANSYS, Abaqus)
- **Deadlines**:

Project	Topic	Statement	Submission
1	Modal analysis	October 12	**November 13**
2	Transient analysis	November 16	**December 18**

Main learning objectives

1. **Understand the fundamentals of dynamic FEA**

   Develop a strong theoretical foundation in the principles of dynamic analysis, including *modal analysis* and *frequency response*.

2. **Analyse the dynamic behaviour of structural components**

   Study the dynamic response of specific structures such as *rods*, *trusses*, *beams*, *plates*, and *shells* under various loading and boundary conditions.

3. **Develop proficiency in finite element formulation for dynamic problems**

   Learn to derive and implement finite element equations for time-dependent structural dynamics problems, including damping and rotational effects.

4. **Apply computational methods for solving dynamic systems**

   Utilise numerical techniques such as direct integration, *modal superposition*, and *time-stepping methods* to solve complex dynamic problems.

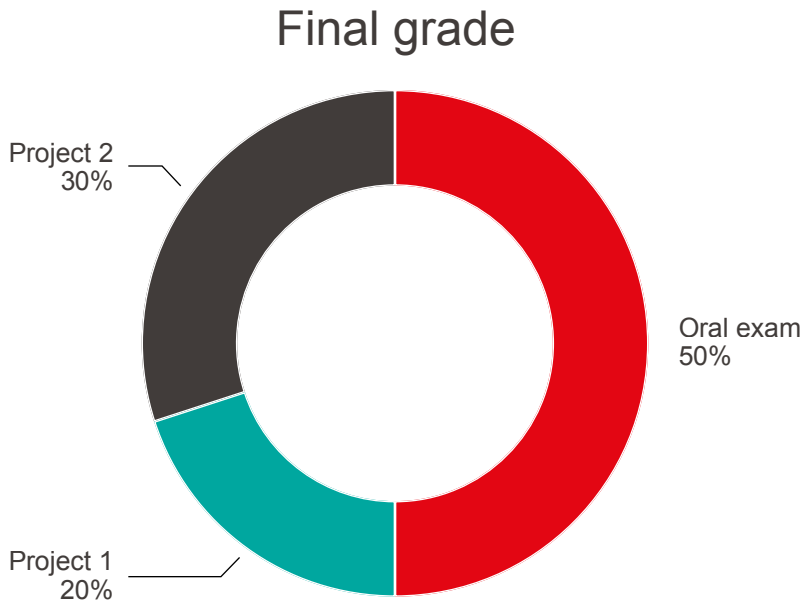
5. **Implement advanced applications in engineering design**

   Apply dynamic FEA to real-world engineering problems, such as vibration analysis and linear dynamic response to dynamic loads, using commercial software.

Assessment method

Final exam

- Fall session
- Oral exam
  - Presentation of mini-project 2
  - Questions about the course



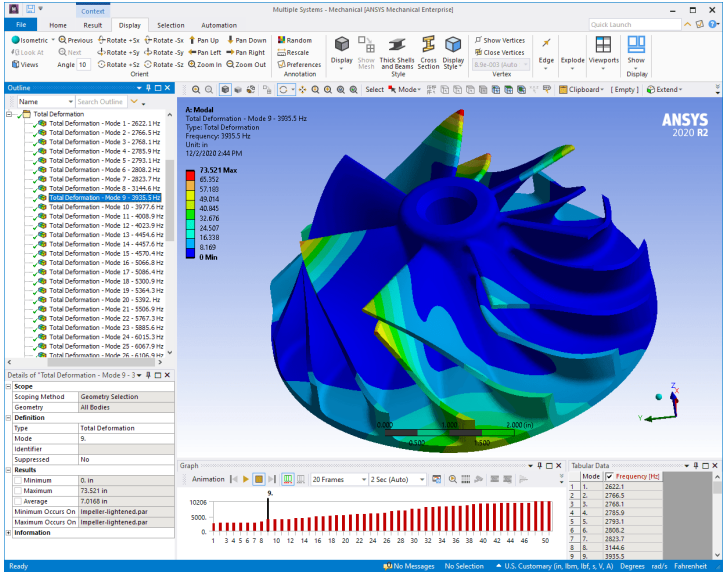
The finite element analysis dichotomy

Code developers

```
35      static public void verify()
36      {
37          cgiSolverBlazeClass solver = new cgiSolverBlazeClass();
38          solver.createStructure();
39
40          StringBuilder sb = new StringBuilder();
41          solver.getDefaultTestPath(ref sb);
42          string sInputFileName = sb.ToString() + "\\tests\\newModels\\Verify-Example4-Sparse.r3a";
43
44          // the following should be called after createStructure()
45          cgiSolverBlazeClass.ListMessageDelegate ListMsg = new cgiSolverBlazeClass.ListMessageDelegate(Callback);
46          cgiSolverBlazeClass.StatusMessageDelegate StatusMsg = new cgiSolverBlazeClass.StatusMessageDelegate(StatusCallback);
47          cgiSolverBlazeClass.SparseSolverProgressDelegate SparseMsg = new cgiSolverBlazeClass.SparseSolverProgressDelegate(MlkProgress);
48          solver.setListMessageFunction(ListMsg);
49          solver.setStatusMessageFunction(StatusMsg);
50          solver.setSparseSolverStatusFunction(SparseMsg);
51
52          solver.setModelType((int)cgiModelEnum.kModel_Frame2D);
53
54          // LENGTH=ft;   DIMENSION=in; FORCE=kip; FORCE_LINE=kip/ft; MOMENT=kip-ft; FORCE_SURFACE=lb/ft^2;
55          // DISPLACEMENT_TRANS=in; DISPLACEMENT_ROTATE=rad; MODULUS=kip/in^2; WEIGHT_DENSITY=lb/ft^3; STRESS=lb/in^2
56          // SPRING_TRANS_1D=lb/in; SPRING_ROTATE_1D=lb-in/rad; SPRING_TRANS_2D=kip/in^2; SPRING_TRANS_3D=kip/in^3
57          solver.setStandardEnglishUnits();
58
59          // define materials
60          List<cgiMaterialCli> listMat = new List<cgiMaterialCli>();
61          cgiMaterialCli mat = new cgiMaterialCli();
62          mat.setId(1);
63          mat.setProperties("Default222", 29000, 0.3, 450);
64          listMat.Add(mat);
65          solver.setMaterials(listMat);
66
67          // define sections
68          List<cgiSectionCli> listSect = new List<cgiSectionCli>();
69          cgiSectionCli sect1 = new cgiSectionCli();
70          sect1.setId(1);
71          sect1.setProperties("W27X84", 24.8, 12.282, 12.7488, 2850, 106, 2.81);
72          listSect.Add(sect1);
73          cgiSectionCli sect2 = new cgiSectionCli();
74          sect2.setId(2);
75          sect2.setProperties("W27X84", 24.8, 12.282, 12.7488, 2850, 106, 2.81);
76          listSect.Add(sect2);
```

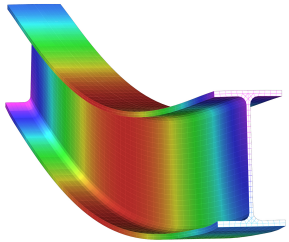
(Source: Cg-inc)

Tool users

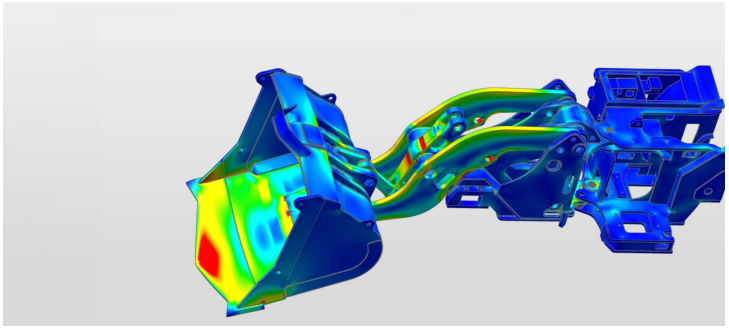


(Source: Padt)

What is static analysis ?



(Source: Ansys)



(Source: Rfamec)

Static analysis determines deformations due to constant loads.

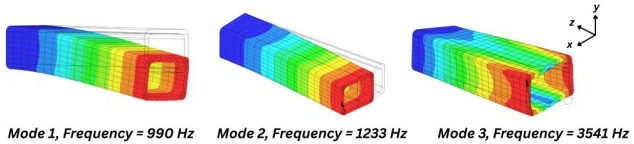
- No matter how you apply the load, the solver assumes it happens infinitely slowly, meaning the loading method has no impact on structural behaviour.
- A constant load doesn't mean a linear response, buckling, yielding, and other nonlinear effects can occur, but the analysis remains static.

What is dynamic analysis ?

The decisive dynamic effect is the presence of inertia forces, forces that arise when the state of motion of a mass changes

Modal analysis studies dynamic properties in the frequency domain.

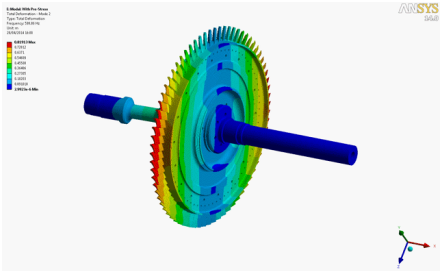
- Natural frequencies are fundamental property in structural design; lower natural frequencies lead to larger displacement and thus greater risk.
- Resonance occurs when external excitation coincides with a natural frequency and can cause catastrophic failure.



(Source: Caeflow)

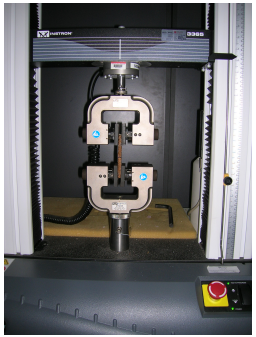
Transient analysis examines time-dependent structural responses to time-dependent excitations.

- Transient analysis helps predict peak stresses, displacements, and vibrations during short-term dynamic events such as impacts, blasts, or sudden load changes.



(Source: Trivista)

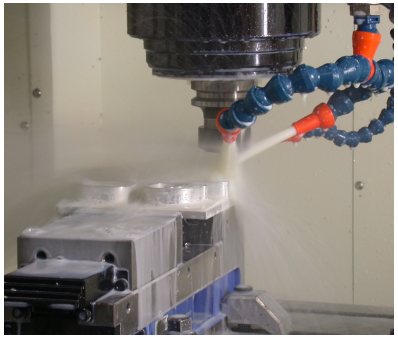
Static or dynamic ?



(Source: Wikipedia)



(Source: Wikipedia)



(Source: Wikipedia)

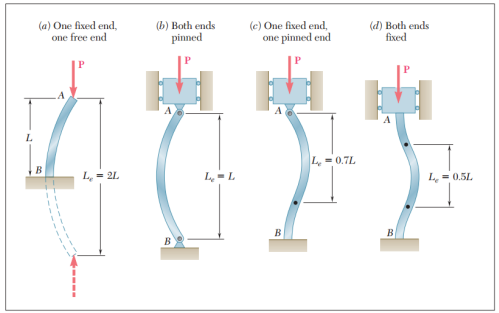


Fig. 10.17 Effective length of column for various end conditions.

(Source: Engineering Stackexchange)



(Source: Wikipedia)

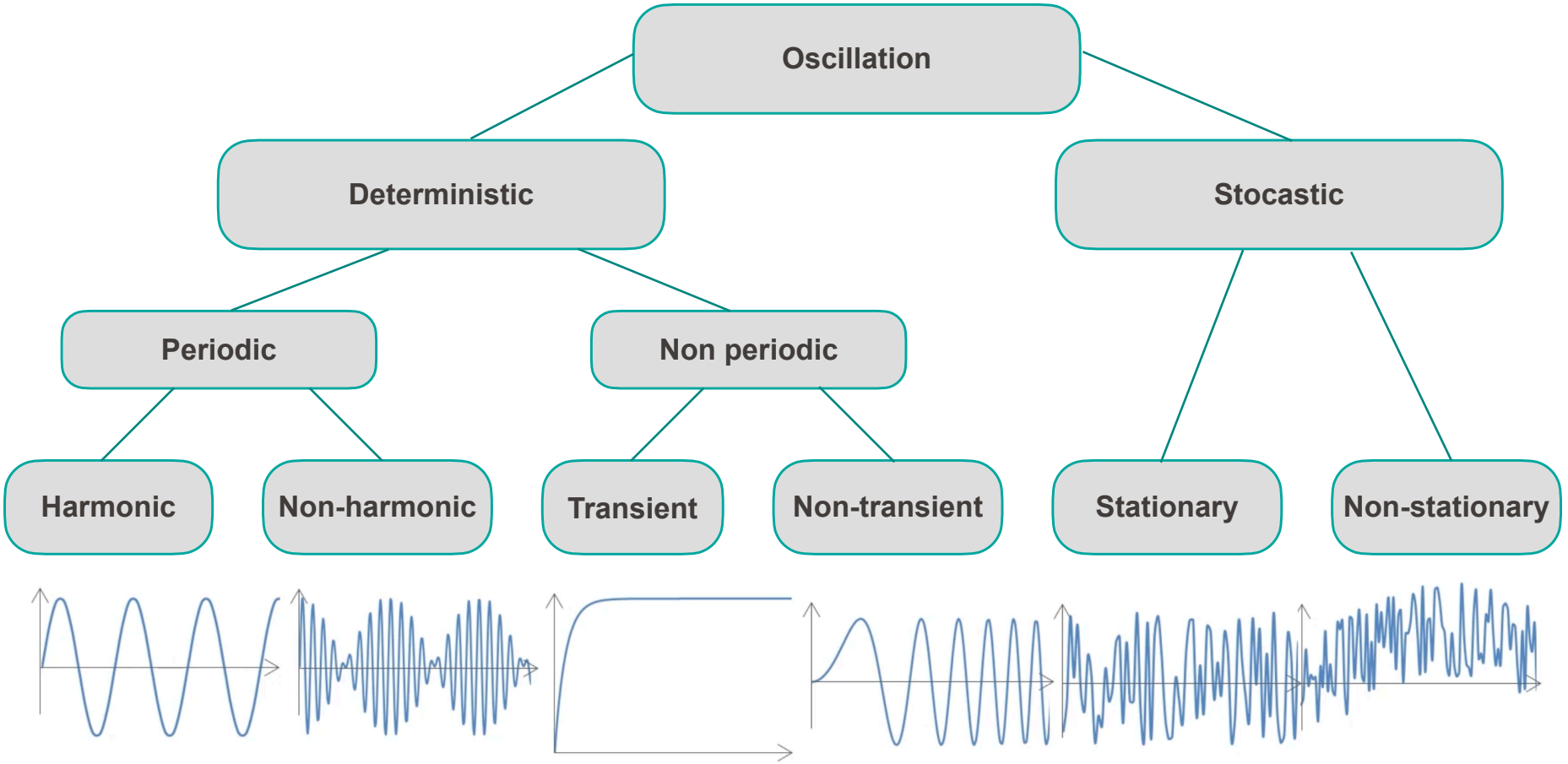


(Source: Wikipedia)



(Source: Wikipedia)

Classification of dynamic analysis



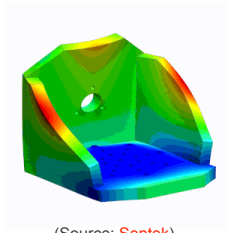
Linear vs non-linear

Key distinction: whether the structural stiffness remains constant as the load changes.

Linear analysis

Constant stiffness

$$K u = F$$



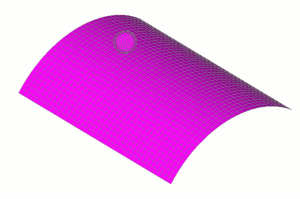
(Source: Sentek)

- Force–displacement relationship is linear.
- Stiffness matrix K is constant during the analysis.
- Typically solved with one linear system for a load case.
- Suitable when deformations and material response remain linear.

Non linear analysis

Stiffness changes along the load path

$$R(u) = F$$



(Source: Enterfea)

- Force–displacement relationship is not a straight line.
- Tangent stiffness K_t varies with the current state.
- Requires incremental / iterative solution procedures.
- Common causes: plasticity, large deformation, and changing contact or element status.



(Source: YouTube)